

Mechanism-Matched Data Scoring: Compression Predicts and Accelerates the Formation of Induction Heads

Luca Columbu

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Abstract

Data selection for language-model pretraining is driven by scalars that blend every capability together: loss under a reference model, similarity to a target distribution, or redundancy. We propose scoring data against a specific *mechanism*: reverse-engineer the circuit a model must build, reimplement its computation as a cheap program, and grade each document by how much it exercises that program. We instantiate the principle for induction heads, whose operation—locate the previous occurrence of the current context and copy what followed—is the operation of the LZ77 compressor. On synthetic induction tasks, per-document compression gain predicts the training step at which induction heads form (Spearman $\rho = -0.98$, three seeds, two architectures), tracks the realised copy length of the data at $\rho = 0.995$ where the generating parameter tracks it at 0.60, and selects a training subset that matches a copy-length oracle exactly. Mechanism tracing shows why: the copy-length distribution selects among *variants* of the induction circuit that differ in attention lag, each with a fixed accuracy ceiling, so plateaus previously read as partial learning are completed circuits of a data-selected variant. A second compressor quantity, normalised compression distance between documents, dissociates from gain and relates to generalisation across name pairs rather than to formation. On real Python code from The Stack, selecting a training subset by gain under a diversity constraint makes induction heads form 1.5–2 $\times$ sooner in attention and 2–3 $\times$ sooner in behaviour than a random subset (three seeds, no overlap), at a stable cost of 0.2 nats of validation loss that doubling the budget does not remove; once formed, both circuits copy rare identifiers equally well in distribution. The compressor’s own match-length histogram predicts, before training, on which corpora selection can act. The complete real-data study cost under ten dollars of compute.

1 Introduction

The data that trains a language model is chosen by heuristics that optimise a single quantity. Perplexity filters keep documents a reference model finds neither too easy nor too hard [Wenzek et al., 2020]; importance resampling matches a target distribution [Xie et al., 2023b]; domain reweighting learns a mixture from a proxy model [Xie et al., 2023a]; learnability-based selection keeps points that reduce held-out loss [Mindermann et al., 2022]; deduplication removes near-copies [Lee et al., 2022, Abbas et al., 2023, Tirumala et al., 2023]. These methods treat the model as a black box. They report averages over benchmarks, and they cannot say which capability a given decision buys or sells, because they never look at what the model builds.

Mechanistic interpretability has meanwhile produced detailed accounts of individual circuits—induction heads [Olsson et al., 2022], indirect-object identification [Wang et al., 2023], modular arithmetic [Nanda et al., 2023]—and has shown that specific distributional properties of training

data switch some of them on and off [Chan et al., 2022, Reddy, 2023]. That work varies the data *generator* and observes the model. It does not provide a way to grade an arbitrary corpus before training.

This paper connects the two. The proposal is simple: **for a circuit the model should build, find the cheapest algorithm that performs the same computation, run it on each document, and use its output as a data score.** We call this *mechanism-matched scoring*, and we instantiate it once, completely, for induction heads. The induction head’s operation—find the earlier occurrence of the current token, attend there, copy what followed—is the operation of the LZ77 compression algorithm [Ziv and Lempel, 1977], whose match statistics are exposed by any `gzip` implementation. The use of compressors to estimate entropy and cross-entropy of strings was developed by Benedetto et al. [2002], Baronchelli et al. [2005]; we use their two quantities—per-document compression gain and cross-document compression distance—and show that they measure two distinct properties of training data with two distinct consequences for the trained model.

Contributions.

1. **Compression gain predicts induction-head formation** (§4.3). Across repeat fractions, three seeds and two architectures, the formation step is a monotone function of gain ($\rho = -0.98$ attention-only, -0.92 with MLPs; -1.0 on condition means). Formation follows the score where the score and the repeated fraction disagree, and gain fails only where the token distribution changes.
2. **The compressor measures the causal variable better than the generator** (§4.4). Gain tracks realised copy length at $\rho = 0.995$; the repeat-fraction parameter that generated the data tracks it at 0.60. Selection by gain matches a copy-length oracle (97% subset overlap, identical formation) and beats selection by the generating parameter in three of three pools.
3. **Copy-length distribution selects the circuit variant** (§4.5). With MLPs present the induction circuit that forms keys on the token k positions back; k is set by the geometry of copies in the data, agrees across seeds, and fixes an exact accuracy ceiling. Plateaus at 0.66–0.76 are completed lag- k circuits. We trace the lag to a positional copier the model builds before any induction head exists.
4. **Compression distance dissociates from gain and relates to generalisation, not formation** (§4.6). The two quantities decorrelate across data knobs (pooled $r = -0.27$); with gain fixed, distance predicts generalisation across held-out name pairs (partial $+0.46$), gain predicts it negatively, and neither predicts when the task circuit forms. Generalisation to held-out roles is governed by a second circuit that in-document repetition trains.
5. **A learned scorer repairs the compressor’s blind spot** (§4.8). Where accidental matches swamp gain (small vocabularies), a reference model with an induction head restores prediction (ρ from -0.72 to -0.93 pooled).
6. **On real code, selection accelerates formation at a measured cost** (§5.2). On 2.5B tokens of Python, a gain-plus-diversity subset forms induction heads $1.5\text{--}2\times$ sooner in attention and $2\text{--}3\times$ sooner in behaviour than a random subset, pays 0.2 nats of validation loss stably across budgets, and yields an equally load-bearing copy head in distribution. The match-length histogram predicts in advance where selection can act: on children’s stories it cannot, and does not.

A methodological thread runs through the paper. Every synthetic task was mechanism-traced *before* it was swept, and this was not optional: the first task we designed to elicit copying was solved by exclusion (§4.7), and the induction task was redesigned three times as tracing exposed positional and fixed-offset shortcuts (Appendix A). A data-selection claim that skips this step is a claim about a mechanism the model may not have built.

2 Related Work

Data selection. Model-based scores—perplexity [Wenzek et al., 2020], learnability [Mindermann et al., 2022], gradient norms [Paul et al., 2021]—and distribution-based scores—DSIR [Xie et al., 2023b], DoReMi [Xie et al., 2023a]—optimise a global quantity. Deduplication [Lee et al., 2022] and its semantic variants [Abbas et al., 2023, Tirumala et al., 2023] remove redundancy and report gains in average loss. Skill-It [Chen et al., 2023] orders data by contribution to evaluation-defined skills. We differ in defining the target as a circuit and the scorer as a reimplementation of that circuit’s computation, and in reporting circuit-level consequences that averages hide.

Compression and language models. Delétang et al. [2024] show that language-model loss is a compression rate under arithmetic coding; Huang et al. [2024] report that compression rate correlates with downstream performance as a global signal; Jiang et al. [2023] classify text with gzip distance. We use compression as a circuit-specific detector and identify the internal statistic—LZ77 match length—that carries the signal.

Induction heads and training data. Olsson et al. [2022] identify induction heads and their abrupt formation; Chan et al. [2022] show that burstiness, Zipfian skew and class count in the data gate in-context learning; Reddy [2023] give a mechanistic model of the dynamics. Those works vary the generator and observe the model. We measure the documents.

3 Method

3.1 Compressor quantities

Following Baronchelli et al. [2005], the LZ77 code rate of a string estimates its entropy, and compressing B after A estimates the cross-entropy $C(A|B) \simeq (L_{A+B} - L_A)/L_B$, where L_X is the compressed length of X . We derive two quantities.

Gain. Per-document compressibility beyond unigram statistics, $\text{gain}(B) = 1 - L_B/L_{\text{shuf}(B)}$, where $\text{shuf}(B)$ is a random token permutation of B (frequencies preserved) and L is the zlib code length in bits. Gain is large when a document contains long verbatim repeats and zero for a document with no structure beyond its token frequencies; single-token repeats and skewed unigram distributions do not count.

Distance. The normalised compression distance $\text{NCD}(A, B) = (L_{AB} - \min(L_A, L_B)) / \max(L_A, L_B)$ [Li et al., 2004]. Corpus diversity is the mean NCD over a random sample of document pairs at fixed length.

Match-length histogram. LZ77 emits a sequence of (offset, length) matches. The distribution of match lengths is the mechanistically meaningful statistic underlying gain, and we use it as a pre-training diagnostic of whether a corpus contains structure the score can act on. We report the share of tokens covered by matches of length ≥ 8 and the mean longest match per document.

All quantities are computed on token-id sequences with a sliding-window LZ77 (zlib), reusing one compressor state for the context when computing conditional scores.

3.2 Why LZ77 matches induction

An induction head [Olsson et al., 2022] composes a previous-token head, which writes “preceded by x ” at each position, with a head that at the current token x attends to positions so labelled and copies the token found there. LZ77 at each position searches backward for the longest string that begins here and has already occurred, and emits a pointer to it. Both locate the previous occurrence of the current context and reuse its continuation. The compressor’s match length is the length of the repeat the induction head can exploit.

3.3 Selection

Given a pool of documents and a token budget, the *gain* selector ranks by gain and takes the top of the ranking. The *gain + diversity* selector proceeds greedily through the ranking, accepting a candidate only if its NCD to each of 16 probe documents drawn from the already-accepted set is at least 0.75, which rejects near-duplicates. We compare against uniform random selection, against a half-gain, half-random *mix*, and, on synthetic data, against oracles that see the generating parameters.

4 Synthetic Experiments

4.1 Tasks

Induction. Random token sequences of length 64 over a vocabulary of 100 tokens, containing a verbatim repeat of a random earlier span at a random offset; the target inside the repeat is the token that followed the earlier occurrence. The copy length is drawn uniformly up to a maximum $\ell_{\max}$ set by the repeat fraction (16–31 across the sweep). Knobs: repeat fraction, vocabulary, noise, number of copies at fixed repeated fraction, and the copy-length distribution (uniform, fixed, geometric, short-tailed). Appendix A records two earlier versions that tracing showed to be solvable without content matching.

Two-name. Template sentences with two names, one duplicated; the target is the non-duplicated name, after Wang et al. [2023]. Knobs: pool size (4–64), Zipf skew (0–2), repetition rate (0–1), noise. Held-out splits: names never used as the answer (held-out IO), name pairs never co-occurring (held-out pairs), and a leak fraction in which held-out names appear as answers. Generalisation is measured on single-sentence probes, because the document-level metric is inflated by in-context repeats.

4.2 Models and measurement

Two-layer four-head transformers ($d_{\text{model}} = 128$) with rotary positions by default; attention-only and with-MLP variants; 3L8H and 4L8H replications. AdamW, learning rate 10^{-3} , warmup 500 steps, no weight decay unless stated. Three seeds per condition; checkpoints every 100 steps.

Formation step. We use three definitions and report their agreement: (i) behavioural, the first checkpoint at which validation accuracy on copied tokens reaches 0.5; (ii) attention-based, the first checkpoint at which any head places mass ≥ 0.5 on the token to be copied at any lag (§4.5); (iii) patching-based, the first checkpoint at which patching the layer-1 heads at target positions recovers

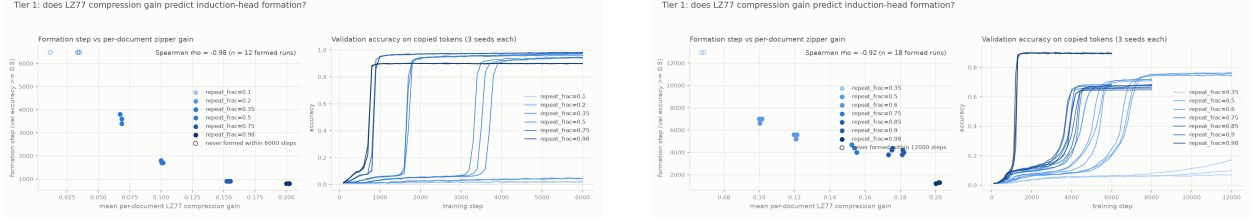


Figure 1: Formation step of the induction head against per-document gain. Attention-only (left): $\rho = -0.98$ over 12 formed runs. With MLPs (right): $\rho = -0.92$ over 18 formed runs. $\rho = -1.0$ on condition means in both. Formation is a jump from accuracy < 0.2 to > 0.8 within one or two 100-step intervals; seed spread 0–200 steps.

Repeat fraction	Gain	Formation, attn-only	Formation, MLP
0.98	0.201	800	1,267
0.90	0.181	n/r	4,000
0.85	0.175	n/r	4,133
0.75	0.154	900	4,367
0.60	0.121	n/r	5,467
0.50	0.101	1,733	6,867
0.35	0.068	3,600	never ($> 12k$)
0.20	0.034	never ($> 6k$)	n/r
0.10	0.012	never ($> 6k$)	n/r

Table 1: Formation step (mean of three seeds) by condition; n/r, not run in that sweep. Seed standard deviations are 0–200 steps (attention-only) and 58–351 steps (MLP).

$\geq 50\%$ of the clean-minus-corrupt logit difference under source-segment corruption, with a 1-nat floor on the effect. Over 39 formed induction runs (i) and (ii) agree at $\rho = +0.99$ with a mean gap of 87 steps, one checkpoint interval; over 45 formed runs (iii) agrees with each at $\rho = +0.96$, and final recovery is 0.97–1.00 everywhere. In attention-only models patching and behaviour coincide (mean gap 70 steps); in MLP models the patched circuit reaches 50% recovery 1,000–2,000 steps before accuracy reaches 0.5, the mechanistic form of the slower “seep” in those models. Correlations below use the behavioural step unless stated.

Circuit analysis. Per checkpoint and per head: attention mass to the relevant positions and direct logit attribution, decomposed exactly into heads, MLPs, embeddings and the LayerNorm-plus-unembed term. Sharpness is the size of the greedy minimal sufficient head set that restores 90% of the logit difference with everything else mean-ablated per position; faithfulness is accuracy with everything outside that set ablated.

4.3 Gain predicts formation

Table 1 lists the repeat-fraction sweep. Below a gain of about 0.07 (attention-only) or 0.10 (MLPs) the circuit does not form within budget. With MLPs, formation is two to four times later throughout and only the densest condition snaps cleanly; the others plateau (§4.5). Weight decay at 10^{-3} and 10^{-2} leaves formation, lag and ceiling unchanged. A 3L8H attention-only replication gives $\rho = -0.94$ over 9 formed runs (formation 2,600 / 1,300 / 933 for repeat fractions 0.5 / 0.75 / 0.98) with the jumps intact.

Two further sweeps separate the score from the repeated fraction. Adding noise lowers gain

[TODO: lag figure: lag and ceiling against copy-length distribution]

Figure 2: With MLPs the induction circuit keys on the token k positions back (lag k). Holding $\ell_{\max} = 24$: fixed-length copies give $k = 0$, formation at 800, accuracy 0.997; uniform lengths give $k = 19$ and a 0.68 ceiling; geometric lengths give $k = 22$ –23, formation at 4,200–4,600 and a 0.59–0.62 ceiling; mostly-short copies never form. All three seeds agree on k in every condition.

and delays formation together ($\rho = -0.92$). Splitting a fixed repeated fraction into 1, 2 or 4 shorter copies lowers gain (0.154, 0.149, 0.125) and delays formation (900, 1,267, 1,367; $\rho = -0.88$): formation follows the score, not the fraction. The one failure is the token distribution: at vocabulary 16 versus 100, gain collapses (0.021 versus 0.101) while formation barely moves (1,367 versus 1,733), the wrong sign. §4.8 repairs this.

4.4 The compressor measures copy length; the generator does not

In Table 1 gain is a deterministic function of the generating parameter, so §4.3 shows only that the compressor orders repetition correctly. A selection experiment separates the two. From a pool of 60,000 documents whose per-document repeat fraction is drawn from $U(0.1, 0.98)$ and hidden, we select 20,000 by (a) gain, (b) the hidden generating parameter, (c) realised copy length measured after generation, (d) loss gain under a fixed reference model (§4.8), and (e) uniformly at random, and train three seeds on each.

Across three independent pools, gain-selected models form the circuit at step 700 in every seed; reference-loss-selected at 700; parameter-selected at 800–1,000; random at 5,200–6,000 or never within budget. The copy-length-selected subset overlaps the gain-selected one by 97% and forms at step 700 in all three seeds with final accuracy 0.94–0.95, the compressor’s numbers exactly. The generating parameter is the worse selector because it describes what was requested rather than produced: the parameter sets a maximum copy length and the realised length is drawn below it, so it tracks realised copy length at $\rho = 0.60$ and admits documents with high nominal repetition but short copies (24% of its subset has copies under 8 tokens; the compressor’s has none). Gain tracks realised copy length at $\rho = 0.995$. All formed models have lag-0 circuits (attention mass 0.82–0.87). [TODO: Figure: formation curves for the five arms.]

4.5 Copy-length distribution selects the circuit variant

A lag- k induction circuit is a layer-0 head carrying the token $k + 1$ positions back and a layer-1 head that matches the current token against it and reads the token k positions after the needed one. It copies correctly whenever the copy offset exceeds k , so its accuracy ceiling is the fraction of targets whose offset exceeds k . The ceiling is observed: on targets with offset above the lag every model scores 0.88–0.93, on targets with offset at or below it 0.01–0.07, and predicted and measured ceilings agree (0.76 / 0.71 at lag 19, 0.83 / 0.73 at lag 12, 0.98 / 0.90 at lag 9–10).

Every MLP model in our sweeps built a lagged variant, with the lag agreeing across seeds in every condition. Across the repeat-fraction sweep the lag was 12, 15, 19, 21 and 22 for $\ell_{\max}$ of 16, 19, 24, 27 and 28, but the lag follows neither the maximum nor coverage: at fixed $\ell_{\max} = 24$ it ranges from 0 to 23 depending only on the copy-length distribution (Figure 2). Attention-only models choose lag 0 at every moderate repeat fraction and lag 8–9 at repeat 0.98 and at vocabulary 16; MLP models never choose lag 0 except on fixed-length data. Weight decay does not alter the lag.

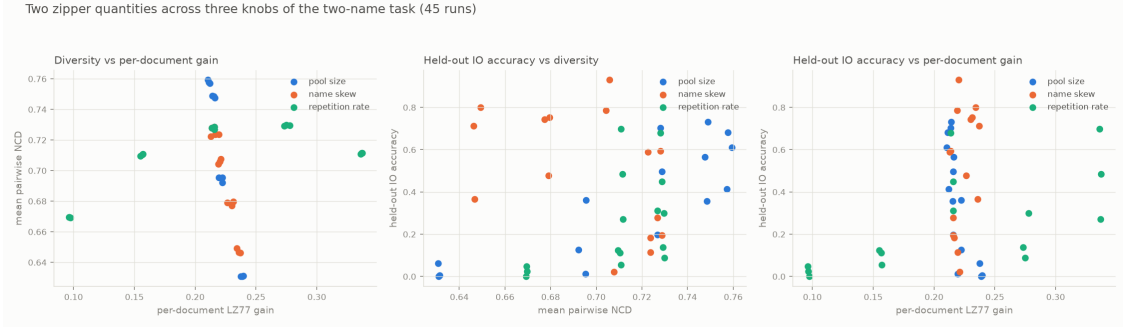


Figure 3: Gain and NCD across the three two-name knobs (45 runs), and each outcome against each quantity.

Where the lag comes from. Relative-offset profiles across checkpoints show that before any induction head exists, the MLP model builds a *positional copier*: heads that attend a fixed d tokens back regardless of content, correct exactly when the copy offset is $d + 1$. The seed sits at the mode of the copy-offset distribution (offset 24 in every dataset here, where accuracy spikes to 0.2–0.5 at offsets 22–25 and is near zero elsewhere at pre-formation checkpoints). The induction head then grows on the seed and inherits its offset: in every run the final lag equals the seed offset minus one, after a drift of one or two positions during formation (uniform: seed 22, lag 19; geometric: seed 23–24, lag 22–23). Fixed-length data yields the textbook lag-0 circuit because its seed lives in layer 1 (all four layer-1 heads at 23 back from step 300) while layer 0 builds previous-token heads; the broader distributions, with less mass at the mode (6–10% of targets against 21%), seed layer 0 instead. So the lag is set by the geometry of where copies sit relative to their sources, not by how much is copied. Two corpora with identical gain but different source-to-copy offset distributions produce different circuits. What decides the seed’s layer is the one untraced step.

Two consequences follow. The plateaus at 0.66–0.76 are not partial circuits but complete lag- k circuits whose ceiling is predicted by k ; a behaviour-only study would misreport them. And the variable that selects the variant is the variable the compressor measures, which is why §4.4 came out as it did.

4.6 Compression distance dissociates from gain and relates to generalisation, not formation

On the two-name task, repetition rate raises gain (0.10 to 0.34) with NCD flat (0.67–0.73), and Zipf skew lowers NCD (0.728 to 0.647) with gain flat (0.215–0.236); pooled with the pool-size sweep the two correlate at $r = -0.27$. Table 2 gives partial Spearman correlations on single-sentence probes at the final checkpoint.

Generalisation to held-out roles is governed instead by the repetition rate of training documents, non-monotonically: probe accuracy on held-out IO names is 0.00, 0.25, 0.40, 0.06, 0.05 at rates 0, 0.25, 0.5, 0.75, 1. In-document repetition trains a content-independent copy mechanism whose promotion of in-context names extends to the held-out ones and counteracts the prior of §4.7, until at high repetition the copy shortcut replaces the task circuit altogether. This is an interaction between the two circuits of this paper, observed on mixed data.

Formation is flat across every knob except the extremes. At pool size 4 the task is memorised; at repetition rate 1.0 every later sentence repeats an earlier one, an induction shortcut pre-empts the task circuit, and formation is delayed from 333 to 1,500 steps: the regime in which data admits

Outcome	NCD (partial)	Gain (partial)
Formation, attention-based	−0.04	+0.03
Formation, patching-based	+0.16	0.00
Held-out pairs, accuracy	+0.46	−0.19
Held-out pairs, logit difference	+0.44	−0.37
Held-out IO, accuracy	+0.19	−0.21
Held-out IO, logit difference	+0.06	−0.07

Table 2: Partial Spearman correlations over 45 two-name runs, each compressor quantity controlling for the other. Diversity relates modestly to generalisation across name pairs and gain negatively; neither predicts formation of the exclusion circuit or generalisation to held-out roles.

a cheaper mechanism and the model builds that instead of the target.

4.7 The two-name circuit is exclusion, not copying

The task was designed to elicit name-copying after Wang et al. [2023]. The model built something else. Two layer-1 heads attend to the duplicated name with mass 0.94–0.99 and push its logit down; no head places more than 0.2 attention on the answer. The answer’s margin of 10–12 logits over pool names absent from the prompt comes from MLP 1 and MLP 0 fed by layer-0 attention, not from a name-mover. Patching the two heads recovers 86–100% of the logit difference; ablating them collapses held-out-IO accuracy to zero in every run and validation accuracy to 0.43–0.51 in five of twelve, a third head serving as backup in the rest. Greedy sufficient sets are 2–5 of 8 heads in 2L4H models (one S-inhibition head plus one or two layer-0 heads) and 7–9 of 32 in a 4L8H replication, spread over three layers with several backup S-heads.

Priors and their cost. Names never used as answers acquire a prior against being the answer, carried by MLP 0 (the unembed-bias term is < 0.1 everywhere). The prior forms *during* circuit formation—the held-out logit difference deepens to about -4.5 over the first 250 leaked examples—then climbs linearly at about 0.01 logits per leaked example and crosses zero after 600–1,000 examples, roughly 1% of sentences seen; a leaked example costs about 800 examples’ worth of erosion before formation and 100 after. Zipf skew weakens the prior faster than it costs pair generalisation (held-out IO 0.36 to 0.63 at the document level; crossing 2,600 to 267 steps).

4.8 A learned scorer repairs the compressor’s blind spot

At vocabulary 16, accidental short matches compress the shuffled baseline as well as the document, and gain collapses (0.021 versus 0.101 at vocabulary 100) while formation does not. A fixed reference model with an induction head, scoring each dataset by its loss gain over shuffled documents, rates vocabularies 16, 32 and 100 at 0.090, 0.099 and 0.104 where gain says 0.021, 0.038 and 0.101. Pooled over 39 induction runs, Spearman with formation is -0.72 for gain and -0.93 for the reference model; the entire gap is the vocabulary family, and within every other family the two agree (Figure 4). Gain and the model’s loss are the same quantity under two compressors [Delétang et al., 2024]; the learned one wins exactly where LZ77’s matching is uninformative. The learned scorer depends on its reference having the circuit: in §4.4 it matched gain document for document only because its reference was the one random-arm seed that had formed.

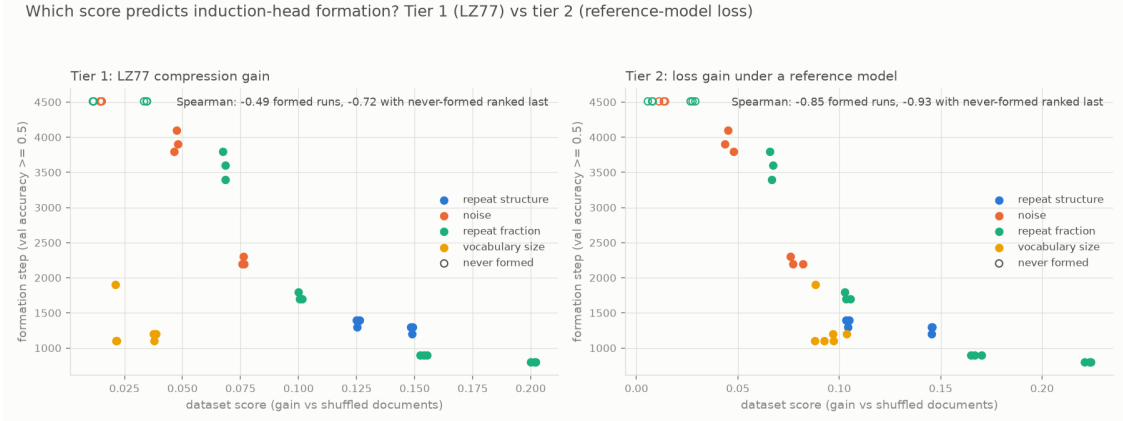


Figure 4: Formation step against the compressor score (left) and against the reference-model score (right), 39 induction runs in four families. The two agree except in the vocabulary family, where the compressor’s score collapses and the model’s does not.

5 Real Corpora

5.1 Children’s stories: the diagnostic predicts a null

On a 65M-token slice of TinyStories no induction head forms in any of seven runs; the prefix-matching score never leaves its initialisation baseline. The match-length histogram says why: median LZ77 match of 2 tokens, 0.1–0.35% of tokens in matches of ≥ 8 , against 27% in the synthetic subset that forms at step 700. Selection moves bigram repeats from 12% to 18% and long matches by tenths of a percent. Natural-prose induction rides on single-token repeats (47% of tokens), which the shuffled-baseline gain discounts, and reported prose phase changes sit at 10^9 – 10^{10} tokens [Olsson et al., 2022], 15–150 times this budget. The score has no range here and the model has no budget here; the histogram identified the first before any model was trained.

5.2 Python code: selection accelerates formation

Setup. The Python split of the deduplicated Stack [Kocetkov et al., 2022], streamed to 2.5B tokens under a word-level vocabulary of 32,768 tokens (8.1% unknown-token share) and cut into 9.77M windows of 256 tokens. Of the pool’s tokens, 26% sit in LZ77 matches of ≥ 8 (mean longest match 28), against 27% in the synthetic subset of §4.4; in the gain-ranked top fifth the share is 62% (mean longest 61), and in the diversity-constrained selection 57% (50). Arms: gain-plus-diversity selection (greedy by gain, $NCD \geq 0.75$ to 16 probes; 247k of the 2.2M windows scanned rejected) and uniform random selection, 1.95M windows each. Model: 4 layers, 8 heads, $d_{\text{model}} = 256$, rotary positions, batch 64, learning rate 10^{-3} , warmup 500, bf16; 30,000 steps (491M tokens, one pass over the arm), three seeds each, checkpoints every 500–1,000 steps; one seed per arm continued to 60,000 steps (983M tokens, a second pass). Measures: the prefix-matching score of Olsson et al. [2022] (best head, top-512 tokens) and induction accuracy on repeated random sequences the model has not seen; validation loss on 20,000 held-out windows; and an in-context delta on rare identifiers (below).

A preliminary local experiment on 64M tokens of installed-package Python, repeated over epochs, grew a head five times stronger in the selected arm but formed in neither; fresh data at the same budget, not more steps, was the fix, as with TinyStories (Appendix B).

Arm	Seed	PM ≥ 0.5	Ind. acc ≥ 0.5	Final PM	Final ind. acc	Val loss
Gain + div.	0	1,000	2,000	0.75	0.86	1.80
Gain + div.	1	1,000	2,000	0.89	0.86	1.80
Gain + div.	2	1,000	2,000	0.83	0.84	1.78
Random	0	2,000	6,000	0.82	0.70	1.60
Random	1	1,500	4,000	0.74	0.78	1.61
Random	2	2,000	5,000	0.74	0.73	1.59
Gain + div., 983M	0	1,000	2,000	0.75	0.86	1.745
Random, 983M	0	2,000	5,500	0.84	0.70	1.557

Table 3: The Stack, 491M tokens unless stated. Formation steps are the first checkpoint at which the measure reaches 0.5; seed 0 has checkpoints every 1,000 steps, seeds 1–2 every 500.

The Stack, Python: two-quantity selection vs random (4L8H, 491M tokens per arm, three seeds; 1B-token continuation dashed)

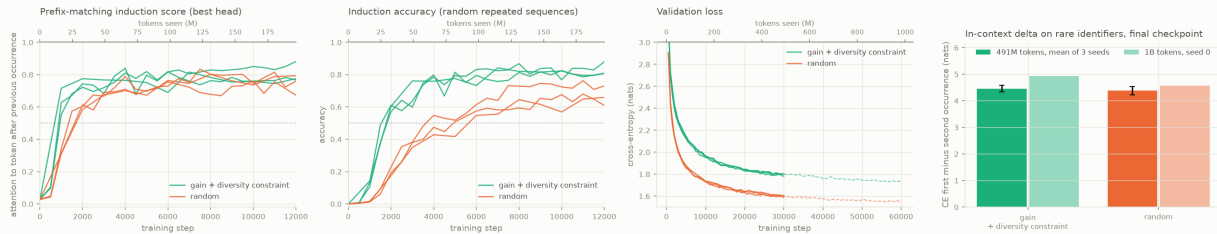


Figure 5: The Stack. Prefix-matching score and induction accuracy over the formation window for both arms and three seeds; validation loss to 983M tokens (dashed: the continued seed); the in-context delta on rare identifiers at the final checkpoint. The selected arm snaps (prefix-matching 0.10 to 0.55–0.72 between steps 500 and 1,000; induction accuracy 0.38–0.49 to 0.57–0.61 between 1,500 and 2,000); the random arm seeps (prefix-matching 0.31 to 0.46–0.61 over steps 1,000–2,000; induction accuracy 0.17–0.23 at 2,000, 0.43–0.55 at 4,000, 0.55–0.62 at 6,000).

Formation. The selected arm forms the head at step 1,000 in every seed and the behaviour at step 2,000 in every seed; the random arm at 1,500–2,000 and 4,000–6,000 (Table 3): 1.5–2× sooner in attention and 2–3× sooner in behaviour, with no overlap on either measure. The selected arm’s induction accuracy stays 0.06–0.16 above the random arm’s to the end of training while prefix-matching overlaps (0.75–0.89 versus 0.74–0.84): both heads attend where they should, and the selected model executes the copy more accurately on random sequences. Doubling the budget leaves formation and final induction accuracy unchanged in both arms and improves validation loss by 0.05 in both; the 0.2-nat gap between arms persists and the loss curves never cross. The trajectories reproduce, between two selections of one real corpus, the snap-versus-seep contrast observed between the attention-only and MLP models of §4.3 (Figure 5).

In-context delta. Absolute loss on repeated identifiers orders like validation loss and so measures distribution quality, not the head. We instead measure, for each rare identifier (not among the 2,000 most frequent names in a 1M-window sample) occurring at least twice in a validation window, the cross-entropy at its first occurrence CE_1 and at its second CE_2 ; the delta $CE_1 - CE_2$ is the in-context benefit, and distribution quality cancels. As controls we compute the same delta for repeated common tokens and a name-swap test in which the identifier is replaced throughout the window’s prefix by a substitute name, reporting the loss moved off the original name, the loss

Arm	Seed	CE ₁	CE ₂	Delta (frac. > 0)	Common-token delta	Prefers swap
Gain + div.	0	9.14	4.56	4.58 (0.86)	0.74	0.57
Gain + div.	1	9.06	4.78	4.28 (0.84)	0.73	0.57
Gain + div.	2	9.07	4.54	4.53 (0.86)	0.73	0.58
Random	0	8.39	3.98	4.41 (0.88)	0.66	0.54
Random	1	8.20	4.02	4.18 (0.87)	0.66	0.47
Random	2	8.29	3.73	4.56 (0.88)	0.66	0.54
Gain + div., 983M	0	9.20	4.27	4.93 (0.86)	0.72	0.60
Random, 983M	0	8.16	3.59	4.56 (0.88)	0.66	0.55

Table 4: In-context delta on 4,807 rare-identifier pairs and 20,867 repeated-common-token pairs from 2,000 validation windows, final checkpoints. Delta declines with occurrence distance in every arm alike (5.1–5.5 nats at ≤ 32 tokens; 2.7–3.0 at 128–256).

moved onto the substitute, and how often the model then prefers the substitute.

The copy head is load-bearing in distribution in both arms: 4.2–4.9 nats on rare identifiers, positive for 84–88% of them, six times the delta on common tokens, and the swap control moves 5–6 nats off the original name and 7.4–8.9 onto the substitute, with the model preferring the substitute about half the time. At the end of training the two arms copy about equally (mean delta 4.46 versus 4.38 over three seeds, within seed spread); CE₁ orders like validation loss, the distribution cost again. At 983M tokens the selected arm’s delta is 4.93 versus 4.56, a modest edge on one seed.

Reading. On real code, selection by the compressor buys *when* the induction circuit forms and a circuit that copies more accurately on syntax-free sequences; it does not buy in-distribution copying once both circuits exist, and it costs 0.2 nats on the distribution that doubling the budget does not recover. The synthetic finding transfers—gain governs formation and a gain-heavy selection pays on the distribution—and the recommendation is unchanged: use both compressor quantities.

Trade-off curve at small scale. On the local 64M-token pool, four arms at matched size—gain only, gain plus diversity, half gain and half random, random—order monotonically on validation loss (2.43, 2.30, 2.13, 2.03) and on the loss at repeated rare identifiers (6.20, 5.68, 5.13, 4.74), while the diversity-constrained arm grows the strongest head of any (prefix-matching 0.10–0.12 in two seeds versus 0.03–0.07 for gain only), and the mix keeps a head twice random’s. Rejecting near-duplicates removes cross-document repetition, which an in-context copy head cannot use, and keeps within-document repetition, which it can. The trade-off is tunable, and the two quantities set its two axes (Figure 6). [TODO: repeat the four arms on The Stack if budget allows.]

6 Discussion

The principle. Gzip predicts induction not because compression measures data quality in general but because LZ77 and the induction head perform the same computation. The generalisation is not “use gzip” but “for each circuit, find the cheapest program that computes what the circuit computes.” Where no such program is at hand, a small model that has already learned the circuit is a learned implementation of it (§4.8).

Python source: one compressor quantity vs two (three seeds per arm, 8,000 steps)

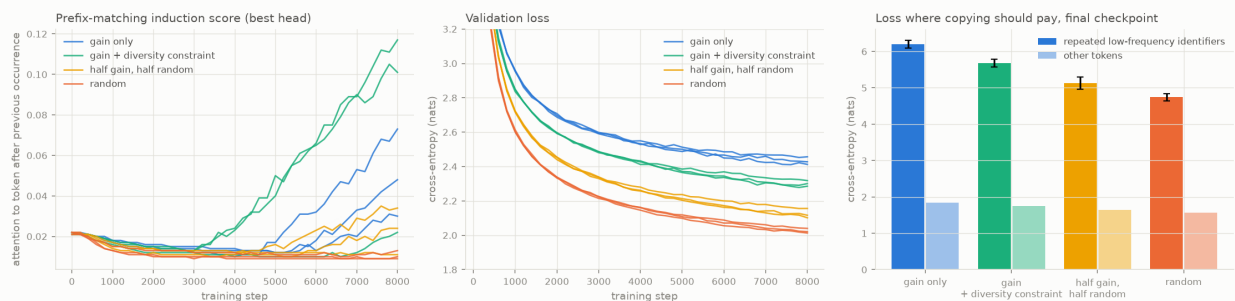


Figure 6: Local Python pool, four arms, three seeds, 8,000 steps: prefix-matching by checkpoint, validation loss, and loss on repeated low-frequency identifiers at the final checkpoint.

What averages hide. Deduplication lowers gain; our results predict, and §5.2 shows the converse of, a delay in induction. Outlier filtering lowers diversity; §4.6 predicts a cost in generalisation across rare combinations. Neither cost appears in a loss average. Mechanism-matched scores make them measurable before training, and the match-length histogram says on which corpora a given score can act at all.

Data picks the variant. That the copy-offset geometry sets a circuit parameter with a built-in ceiling (§4.5), that in-document repetition trains a second circuit whose reach over held-out names rises and then collapses (§4.6), and that two selections of one real corpus yield circuits that copy equally in distribution but unequally on syntax-free sequences (§5.2), suggest a broader question: which data statistics set which circuit parameters. We have one traced law and one real-data echo of it.

Limitations. Models of 2–4 layers; one circuit family; one learning rate. The real-data acceleration is measured at formation steps of one to six thousand, early relative to any production budget; the persistent difference is in the circuit’s generality, not its in-distribution use. The lag mechanism is traced to the positional seed but the choice of the seed’s layer is not explained. The exclusion circuit’s formation is insensitive to every data property varied, so the two-name task tests generalisation, not formation. Prose corpora at our budget are below the induction phase change, and our diagnostic correctly reports that the score has no range there, but we have not shown selection acting on prose. The 0.2-nat distribution gap is a fixed cost of selecting a fifth of a pool by any compression criterion and would need a mix or a curriculum to remove.

Future work. A second circuit where the hand-written scorer should fail and the learned one should win: synthetic facts with paraphrases, where gain selects against the data that builds retrieval and NCD selects for it. What decides the seed’s layer. The four-arm trade-off on The Stack. Interference between circuits on mixed data, of which the repetition-rate result is a first instance.

Reproducibility. All synthetic experiments run on a laptop CPU or its integrated GPU. The complete real-code study—data download, tokenisation, scoring, eight training runs to 491M–983M tokens, and analysis—cost about \$6.60 of rented GPU and CPU time plus a network volume. Code, findings log and this draft: <https://github.com/lucacolumbu/LLM-research>.

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A Induction-task redesigns

Three versions were built; each was mechanism-traced before sweeping. (1) Fixed repeat offset: solved by a layer-0 positional head with no content matching. (2) Random offset, fixed length: content matching present, but keyed on “the token r positions back” because a fixed copy length privileges that feature. (3) Random offset and length: no shortcut remains; used throughout. Learned absolute positions with a short warmup, not weight decay, were found to block formation in an early MLP configuration; weight decay of 0, 10^{-3} and 10^{-2} gives identical formation, lag and ceiling.

B Local code pool

12,037 installed-package Python files, 498k windows of 128 tokens; top 100k windows by gain versus three random 100k subsets; 2L4H, 8,000 and 24,000 steps. Gain of subset 0.365 versus 0.162; 45% versus 14% of tokens in matches ≥ 8 (mean longest match 30 versus 13). Prefix-matching at 8,000 steps 0.03–0.07 (rising from step 5,000) versus 0.01 flat; at 24,000 steps 0.17 versus 0.03; validation loss 2.41 versus 1.90. In-context delta on rare identifiers 1.7–2.0 nats in every arm, name-swap control 1.3–1.8 nats, with no ordering by arm: at this budget the accelerated head was not yet load-bearing. Four-arm results in §5.2.

C Additional figures

[TODO: per-run patching summaries; tier comparison per family.]

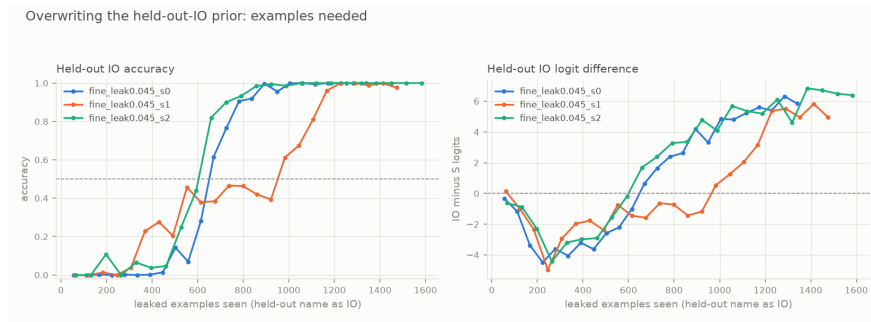


Figure 7: Held-out IO accuracy against leaked examples seen, two-name task.